

Among-population disequilibrium between unlinked ancestry clusters tracks a shared ancestry axis, not pair-specific incompatibility

Draft results summary (Module D, the intrinsic arm). Methods are deferred to Supplementary Materials. Findings are descriptive: the screen characterises the observed distribution of among-population two-locus disequilibrium and ranks candidate pairs, but a pair-specific incompatibility is licensed only as an excess over the forthcoming recombination-matched neutral null. All statements below concern that observed distribution.

Results

Design. An intrinsic incompatibility between two loci acts on their joint parental ancestry, and across independent replicate populations should leave a pair-level trace that per-locus sorting cannot show: the two loci fixing compatible ancestry in the same direction repeatedly, so that their allele frequencies covary across populations. Between loci that recombine freely this covariance cannot be produced by linkage, and under drift alone independent replicates sort such loci independently, so its neutral expectation is near zero. We define two clusters as *unlinked* on the genetic map—on different chromosomes, or on the same chromosome more than 10 cM apart—because admixture linkage disequilibrium decays as $(1 - c)^t$ irrespective of physical distance, and at the ~ 50 generations of hybridisation implied for these populations 10 cM corresponds to $\sim 99\%$ decay (and to roughly five mean ancestry-tract lengths). We screened all such unlinked pairs of clusters that were polymorphic in the parents (pooled parental MAF ≥ 0.15 ; 27,223 clusters, $> 3.6 \times 10^8$ pairs), measuring the among-population covariance as R_{st} , the correlation across the twenty populations of a cluster’s mean consensus dosage, and evaluating the exact Ohta decomposition on the 401,945 pairs reaching $|R_{\text{st}}| \geq 0.7$. Same-chromosome pairs within 10 cM (about 1.8 million) are retained as a close-linkage positive control.

Linkage raises among-population disequilibrium, and it decays to the unlinked baseline by ~ 10 cM. The close-linked control behaves as it must: same-chromosome pairs within 10 cM carry a heavier tail of R_{st} than unlinked pairs, exceeding them by $1.8\times$, $2.9\times$ and $16\times$ at $R_{\text{st}} \geq 0.5$, 0.7 and 0.9 . Resolving this by genetic distance shows the expected decay: mean among-population R_{st}^2 between same-chromosome clusters falls from 0.082 at ~ 1 cM to the inter-chromosomal baseline of 0.058 by about 10–15 cM (0.061 at 8–10 cM). The empirical decorrelation scale therefore matches the ~ 50 -generation expectation, and confirms from the data that clusters more than 10 cM apart carry no more among-population disequilibrium than clusters on different chromosomes.

Among unlinked pairs, the decomposition points to structure, not epistasis. On the high- R_{st} unlinked pairs the Ohta decomposition is unambiguous about the source of the disequilibrium. The among-population component exceeds the within-population component in virtually every pair (median $D_{ST}^2 = 0.13$ versus $D_{IS}^2 = 0.005$; $D_{ST}^2 > D_{IS}^2$ in $> 99.99\%$ of pairs), while the disequilibrium of the averaged population is negligible (median $D_{ST}^2 = 0.005$, below D_{IS}^2 in every pair). A large among-population variance component together with a vanishing average disequilibrium is the classic signature of correlated allele-frequency *divergence* among subpopulations—drift and shared structure—rather than of a systematic two-locus association of the kind epistatic selection would maintain within populations.

A single shared ancestry axis accounts for the pattern. The covariance is predominantly positive: 62% of the high- R_{st} unlinked pairs covary in the same direction, and among pairs in which both clusters are themselves directionally sorted, same-direction pairs outnumber opposite-direction pairs 1.3 to 1. This is what a single genome-wide ancestry gradient produces—populations that fixed *F. aquilonia* ancestry did so across many diagnostic loci at once, so those loci covary positively regardless of any interaction between them. The among-population disequilibrium between unlinked clusters is thus dominated by the same directional ancestry sorting characterised in the per-locus analyses, now seen as the covariance it induces between loci, rather than by pair-specific coupling.

Interpretation and outlook. Before a neutral null, the screen finds no evidence of pair-specific incompatibility. Linkage raises among-population disequilibrium and it decays away by ~ 10 cM; beyond that distance the disequilibrium that remains reads, by its own decomposition, as correlated ancestry divergence along one genome-wide axis. This is the expected descriptive outcome, and it sharpens rather than answers the intrinsic question. A Dobzhansky–Muller interaction would appear as an *excess* of among-population disequilibrium, at specific unlinked pairs, over what neutral sorting of the same replicate populations produces. Module D is therefore run as a distribution-generator: it emits the observed R_{st} and among-population disequilibrium distributions, and candidate incompatibilities are defined downstream as the pairs exceeding the recombination-matched neutral null, evaluated by applying the identical pre-filter and decomposition to simulated genotypes. That comparison, with the concurrent close-linkage reference, is the decisive next step.